

Class Bivalvia

Bivalves

Occurrence c.15,000 spp. worldwide; in aquatic habitats, mostly on the sea bed, but some in freshwater



Bivalves are the only mollusks that have a shell consisting of 2 parts, called valves, hinged together. In most cases, the valves can be closed tightly to protect all or a large part of the animal's body. Mussels, clams, oysters, and scallops are among the most familiar bivalves, providing a valuable source of food for humans as well as other creatures. Some species, such as the piddock and shipworm, have small, sharp-edged valves, used like drill bits to bore through wood or soft rock. Sizes

of bivalve mollusks vary greatly, from less than $\frac{1}{16}$ in (2 mm) to as much as $3\frac{1}{4}$ ft (1 m) across. The largest species (giant clam) is about 2 billion times heavier than the smallest. Bivalves are usually static, although some—for example, scallops—can swim. Most are filter feeders, using modified gills to strain food from the water current. The water is drawn in and pumped out through tubes or siphons—especially important for species that live buried in mud or sand. The siphons may be so large as to be permanently outside the shell. Bivalves have tiny heads, and they do not have a radula—the mouthpart that other mollusks use to rasp away at their food. The sexes are separate, although some bivalves may be hermaphrodite.



MUSSELS

Mussels attach themselves to hard surfaces by means of small threads, allowing them to hold fast against wave action and water currents. They often live in large clusters, and can cause problems for coastal power stations by blocking their cooling systems. Mussels are often raised in aquaculture systems. *Mytilus edulis* (above), a species found in northern waters, may be 6 in (15 cm) long.



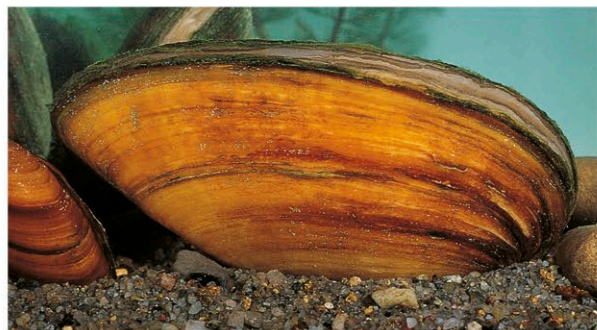
SADDLE OYSTERS

Like true oysters, saddle oysters or jingle shells live with one valve attached to a hard surface. The method of attachment is peculiar: a calcified plug passes from the upper valve through a hole in the lower one onto the surface. The Atlantic *Anomia ephippium* (above) is about $2\frac{1}{4}$ in (6 cm) long.



GIANT CLAMS

The tropical, reef-dwelling giant clam, *Tridacna* sp. (above), is the largest bivalve; it may measure more than $3\frac{1}{4}$ ft (1 m) across and weigh over 440 lb (200 kg). Contrary to popular myth, it is harmless to humans. As with all giant clams, most of its dietary needs are met by symbiotic algae that live in its fleshy mantle; using sunlight, the algae produce a supply of energy-rich food. The clam also filters microscopic particles of organic matter from the sea.



SWAN MUSSELS

Most bivalves are marine, but some—such as this swan mussel—inhabit freshwater. A species of the genus *Anodonta*, it is found in North America and Eurasia, and can grow to 9 in (23 cm) across. Their shells are thin, mainly because calcium is less readily available in freshwater. In their larval stages, they are parasitic on fishes.



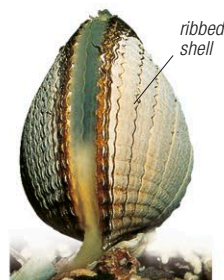
SCALLOPS

Unusually among bivalve mollusks, scallops can swim using a form of water-jet propulsion, to which they usually resort to escape predatory starfish. This North Atlantic species, *Aequipecten opercularis*, may attain a diameter of $3\frac{1}{2}$ in (9 cm).



THORNY OYSTERS

The tropical thorny oysters, found mostly on rocky outcrops and coral reefs, grow up to 8 in (20 cm) in diameter and have shells covered with elongated spines, as in the *Spondylus* species pictured above. Their shells often become covered with algae, which may serve as camouflage.



COCKLES

Cockles live in sandy sediments, burrowing just below the surface so the body is half in and half out of the sand. The shell is usually strongly ribbed, stout, and semispherical. Several species, including the Atlantic *Cerastoderma edule* (above)—which is 2 in (5 cm) high—are harvested commercially.



EDIBLE OYSTERS

The edible oyster cements its left-hand valve directly onto a surface such as a rock or shell, so that the animal lives lying on its side. A valued food source since prehistoric times, oysters are among the most popular of edible mollusks. This North Atlantic *Ostrea* species may grow up to 4 in (10 cm) in diameter.

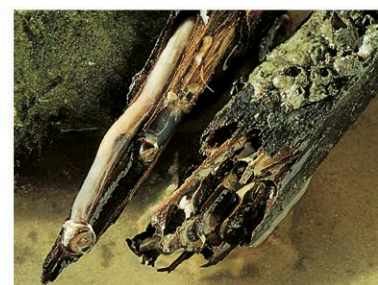


RAZOR CLAMS

Razor clams are so-called because of their resemblance to closed "cut-throat" razors. Occurring in almost all latitudes, they inhabit deep, vertical burrows in sandy sediments, using the shell valves to wedge themselves in. The species whose shell is shown here, *Ensis siliqua*, may grow up to 10 in (25 cm) long.

PIDDOCKS

Using their elliptical shells like drill bits, piddocks bore into materials such as rock, wood, peat, and clay, and live in the excavated burrow. Some piddocks are phosphorescent, glowing with a ghostly blue-green light at night. This North Atlantic species, *Pholas dactylus* (left), may be up to 6 in (15 cm) across.



SHIPWORMS

These wood-boring bivalves occur in all seas, but are particularly common in warm waters. The relatively small shell is used to gouge out a tunnel in which to live, and the shipworm lines it with a chalky tube. Their bodies may extend up to $6\frac{1}{2}$ ft (2 m), although the common shipworm, *Teredo navalis* (pictured), rarely exceeds a tenth of this size.